



M002 Week 8 — English Worksheet (Beginner)

Topic: Capstone — 20×20×20 Cube Maze · **Course:** Maze Madness · **Level:** Beginner · **Time:** about 45 minutes

This is the **final project**. You design your own maze with a redstone door and a lever, then write code that solves it.

1 · Predict 🧠

Read the steps. Before you imagine the agent doing it, circle or write your answer.

Your maze needs:

- walls
- 1 redstone door
- 1 lever
- a start and a finish

Sketch (in words) your maze. Where is the start? Where is the goal? Where is the lever?


```
keep doing until at goal:
    if wall ahead:
        turn right
    otherwise if lever ahead:
        interact ahead
    otherwise:
        move forward
```

The agent reaches the lever. What does it do? Circle one: turn right · interact · move forward

2 · Find the Difference 🐛

Each pair shows clean steps first, then a broken version of the same idea. Circle what's different and write one short sentence about the bug.

Pair A — The agent should **interact** only when there is a **lever** ahead.

```
# clean
if lever ahead:
    interact ahead
otherwise:
    move forward
```

```
# buggy
interact ahead
move forward
```

What is wrong?

Pair B — The agent should **stop** when it reaches the goal.

```
# clean
keep doing until at goal:
    if wall ahead:
        turn right
    otherwise:
        move forward
```

```
# buggy
keep doing forever:
    if wall ahead:
        turn right
    otherwise:
        move forward
```

What is wrong? When does the buggy agent stop?

3 · Fill the Gap

The agent should keep going **until it reaches the goal**. One part is missing. Fill it in using the word bank.

```
keep doing until _____:
  if wall ahead:
    turn right
  otherwise:
    move forward
```

Word bank: at goal · forever · wall ahead

Write the missing part:

4 · Tell Me What You Built 📷

Now build your **final project**. Your maze needs:

- a **20 × 20 × 20** cube shape
- walls and a clear path
- **1 redstone door**
- **1 lever** the agent must flip
- a clear **start** and **finish**
- agent code that solves it using `if/otherwise` and `interact`

When you finish, come back here. Send a photo or video of the agent solving your maze, then explain what you did. Use these sentence starters — write 2 or 3 sentences.

My maze has ...

The agent's plan is ...

The thing I am most proud of is ...

5 · Record Your Walkthrough 🎥

Take a video on your phone (or a parent's phone) while the agent runs your maze. Talk like you are teaching a friend. Try to use these words: **maze, lever, door, goal**.

1. Show the maze, the start, the door, and the lever.
2. Run your code and follow the agent to the goal.
3. Say what you learned in this course.

Write what you will say in your video. You can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk. This is the **last week** of Maze Madness — congratulations!